# Business Statistics

Module 1
Video 1.2
Statistics

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# Agenda

- The seven characteristics of statistics with examples

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# Characteristics of Statistics

## 1. Aggregate of Facts

- "Aggregate" - Cannot be a single numerical value
- "Facts" - Only something that can be measured
- Examples
  - Height of a student is 5 feet 4 inches ✗
  - Heights of students of a certain class ✓

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# Characteristics of Statistics

## 2. Affected by Several Causes

- A statistic is affected by not a single but multiple causes
- This implies that we cannot study the effect of a particular factor in isolation
- Example
  - Sales of the latest model of an iPhone
  - Can be affected by the number of new features, price, supply, marketing, etc.

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# Characteristics of Statistics

## 3. Numerically Expressed

- A statistic should take a quantitative form and expressed as a numerical figure
- This implies that qualitative phenomenon cannot be statistics
- Example
  - Happiness of employees in an organization ✗
  - Number of employees in an organization who say they are happy ✓

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# Characteristics of Statistics

## 4. Enumerated or Estimated as Accurately as Possible

- Data collected by explicit counting- this will be exact
  - For example, the height of students in a class
  - Cannot leave out a single student
- Data collected by estimation when counting is not possible- accuracy varies
  - For example, average height of children aged 12-15 years in India
  - We cannot count every child in the age group thus we estimate using samples
  - Leaving out 10 children in the age group will not significantly affect the estimate

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# Characteristics of Statistics

## 5. Collected in a Systematic Manner

- Unmethodical and disorganized data collection is undesirable
- It hampers the accuracy of the measurement
- It may lead to us making wrong conclusions

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# Characteristics of Statistics

## 6. Collected for a Planned Purpose

- Before starting the process of data collection, the purpose should be clearly defined
- Resources and time will be wasted in collecting irrelevant data
- The relevant data may not be collected at all or its quality may be affected
- Example
  - We want to measure the number of Non-fiction books sold by a bookstore
  - Counting the number of fiction and comic books is irrelevant

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# Characteristics of Statistics

## 7. Comparability

- Bing able to compare different statistics or over is what makes them useful
- Comparability is essential for making inferences, establishing or detecting changes in trends
- Example
  - We measure the heights of students from two different classes
  - We can then compare the  average height of each of the two

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# Summary

1. Aggregate of facts
2. Affected by several causes
3. Numerically expressed
4. Enumerated or estimated as accurately as possible
5. Collected in a systematic manner
6. Collected for a planned purpose
7. Comparability

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